Linear Algebra (MT-1004)

Homework#04

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Question 1

Let $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $y = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$, and $z = \begin{bmatrix} 0 \\ 4 \\ 8 \end{bmatrix}$.
Is the set $\{x, y, z\}$ linearly dependent?

Question 2

Let the vectors be:

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad y = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \quad z = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Use Python's `sympy` library to find the condition(s) on a, b, c for which the set $\{x, y, z\}$ is linearly dependent.

Instructions:

1. Define a, b, c as symbolic variables using `symbols()`.
2. Represent vectors x, y, z as `Matrix` objects.
3. Form a matrix with x, y, z as columns.
4. Use the `det()` or `rank()` method to find a condition for linear dependence.
5. Solve the condition symbolically using `solve()` to find a relationship among a, b, c.